

Karthik Prasad

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EDUCATION

Illinois Mathematics and Science Academy (IMSA)

High School Diploma

Aurora, IL

Aug. 2023 – Present

RESEARCH

MIT PRIMES-USA Researcher

MIT, Indiana University-Bloomington

Jan 2025 – Present

Remote

- Classified Yetter-Drinfeld Modules over the Pansera Algebras under Julia Plavnik at IU-Bloomington.
- Presented at the 2026 Joint Mathematics Meeting, won Top 300 in Regeneron Science Talent Search.
- Expect to submit for publication in Spring 2026.

Theoretical Physics Researcher

Illinois Mathematics and Science Academy

Aug. 2024 – Present

Aurora, IL

- Studied gauge theory, quantum field theory, and generalized symmetries.
- Computed the 0-form symmetries of the $R_{2,2n}$ twisted A_{2n} theory.
- Presenting at American Physical Society Global Summit 2026.

IMSA-CMS Particle Physics Researcher

IMSA, Fermilab

Jun 2025 – Present

Batavia, IL

- Implemented leptonic and hadronic taus for doubly-charged Higgs search.
- Worked with C++, Combine, and ROOT, using a remote Fermilab Linux server, CMS data, and Github.
- Undergraduate-level work; independence of a graduate student.

Research Assistant

Beam Alpha Incorporated

Jun 2025 – Present

West Chicago, IL

- Independently designed+constructed experiment to measure adhesion fractions of water on Zirconia.
- Taught junior interns machining, electrical, coding, and data techniques.

AWARDS

Regeneron Science Talent Search Semifinalist: Top 300 out of 2618 competitors in America's oldest science fair.
MIT THINK Semifinalist: Top 2% of 600+ submissions; won for a proposal to counteract resonance in 3D printing.
United States Physics Olympiad Semifinalist: 518 out of 5848 contestants, received rank of Distinction on exam.
Life Scout: Boy Scouts, expected *Eagle Scout* Spring 2026.

TALKS

On Classifying Yetter-Drinfeld Modules over the Pansera Algebras

15th MIT PRIMES Conference

Oct. 19, 2025

On Classifying Yetter-Drinfeld Modules over the Pansera Algebras

Joint Mathematics Meetings 2026

Jan. 6, 2026

Global Form of Flavor Symmetry Groups in Twisted A_{2N} Class S Theories

IMSAloquium 2025

Apr. 22, 2025

Global Form of Flavor Symmetry Groups in Twisted A_{2N} Class S Theories

APS Global Summit

Mar. 16, 2026

Galois Theory in a Single Night^E

Ross Mathematics Program

July 20–21, 2025

Some Perspectives on p -adic Numbers^E

APEX-Maths in Algebraic Geometry

July 28, 2025

Point-Counting on Varieties^E

APEX-Maths in Algebraic Geometry

Aug. 4, 2025

^E denotes an expository talk.

TEACHING

- Simple Science** | *Hindu Temple of Greater Chicago* Aug 2019 - Present
- Taught demonstration-based science to elementary schoolers, covering topics in simple machines, optics, renewable energy, automobiles, stars, and more.
- Quantum Mechanics** | *Ross/Ohio 2025* Jun2025 – Present
- Taught an introduction to quantum mechanics from the perspective of mathematical symmetry (e.g representations of Lie groups and Lie algebras); ended by deriving the representation theory of $SU(2)$
 - Roughly followed Peter Woit's *Groups, Representations, and Quantum Theory* through Chapter 9.
- Galois Theory in a Single Night** | *Ross/Ohio 2025* 7/20/2025
- Taught an overnight lecture in Galois theory; went from basics of abstract algebra to a galois-theoretic proof of quadratic reciprocity

EXTRACURRICULAR ACTIVITIES

- Hadron** | *Illinois Mathematics and Science Academy* Aug 2023 – Present
- Managed a team of writers on producing quality scientific journalism for IMSA.
 - Led team to produce 5-15 science articles/month; personally edited 5 articles a month; trained 10 new writers a semester; organized bimonthly speaker events.
- Ross Mathematics Program** | *Ross Mathematics Foundation* Summer 2024, Summer 2025
- In 2024, first-year participant; solved twenty number theory problem sets; took courses on knot theory and advanced number theory.
 - In 2025, junior counselor; mentored 4 1st-years; designed and taught a 6-week senior-level quantum mechanics course.
- APEX-Maths Program in Algebraic Geometry** | *APEX Learning* Summer 2025
- Solved problem sets in senior-level abstract algebraic geometry and commutative algebra in Romania.
 - One of 2 out of 50 students from 6 countries to give talks.
- Boy Scouts** | *Scouts America Troop 107* Aug 2018 – Present
- Independently led 100-man-hour service project; held 2nd highest youth leadership position in 20-scout troop; individually maintained troop website.
 - High Adventures in the Canada-USA Boundary Waters, Philmont Scout Ranch, and the Florida Keys.
 - Rank of Life Scout; expected Eagle Scout early 2026.

RELEVANT COURSEWORK

- Illinois Mathematics and Science Academy** | *General Studies* Aug 2023 – Present
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| <ul style="list-style-type: none">• Point-Set Topology• Abstract Algebra• Theory of Analysis• Complex Analysis[†]• Linear Algebra• Number Theory• Calculus-Based Electricity and Magnetism | <ul style="list-style-type: none">• Modern Physics• Astrophysics[†]• The Physical Chemistry of Materials• Organic Chemistry I• Organic Chemistry II• CS Seminar: Linux and Cybersecurity |
|---|--|
- Ross/Ohio 2024** | *Mathematics* Jun 2024 – Aug 2024
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| <ul style="list-style-type: none">• First-Year Number Theory[†]• Quadratic Reciprocity and its Descendants[†] | <ul style="list-style-type: none">• Knot Theory[†]• Fusion Categories[†] |
|--|---|
- Ross/Ohio 2025** | *Mathematics* Jun 2025 – Aug 2025
- Arithmetic of Curves[†]
- Apex-Maths in Algebraic Geometry** | *Mathematics* Jul 2025 – Aug 2025

- Introduction to Algebraic Geometry[†]

North Central College | *General Studies*

Aug 2022 – May 2023

- Mechanics and Heat
- Introduction to Computer Programming

[†] indicates a course was not taken for a letter grade, due to being an independent study or an ungraded course.

SKILLS

Programming Languages, Packages, Tools: Java, Python, C/C++, Github, Linux, Combine, ROOT

Mathematics: Undergraduate math curriculum, experience with multiple math-heavy research projects.

Scientific Writing: Experience with scientific writing for technical and non-technical audiences.